

The Journey of Building ”A Trip On The Bloch Sphere”

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Latest Update: June 17, 2026

1 Introduction

This document chronicles my journey in creating **A Trip on The Bloch Sphere**. It captures my thought process, all my fails, successes, and ideas for the future to come.

2 The Math

Since finding the probability amplitudes of a qubit without explicit calculation is impossible, it is necessary to use matrix multiplication to find the state after each operation applied. While I could use Python or a library that already implements matrix multiplication, what’s the fun in that?

So the first step is to make a library that implements matrix multiplication, and what better language to do it in than C? It’s fast, simple, and straightforward.

It has all I need: no more, no less.

The first step is to use a struct to implement complex numbers.